MT-LNN: A Microtubule-Inspired Liquid Neural Architecture with Global Workspace Theory Bottleneck and Anesthesia Validation

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Abstract

We introduce **MT-LNN** (**M**icro**t**ubule-Inspired **L**iquid **N**eural **N**etwork), a neural architecture that bridges three lines of scientific theory: the computational role of neuronal microtubules (MTs), Closed-form Liquid Time-Constant Networks (CfLTC), and the Global Workspace Theory (GWT) of consciousness. MT-LNN replaces the Transformer FFN with a *Microtubule Dynamic Layer* (MT-DL) — 13 parallel CfLTC channels with multi-scale resonance ($P \times S$ banks, $S = 5$ geometric τ sweep), three-way lateral coupling (static identity + synchronous nearest-neighbour via `torch.roll` + content-aware RMC attention), periodic GTP-cap renewal, and MAP-protein stabilisation gates. A *Microtubule Attention* module fuses scalar and optional low-rank bilinear polarity bias with a per-head ALiBi-style GTP-decay gate ($64\times$ receptive-field spread), implemented over GQA with Flash-Attention. A *Global Workspace Theory Bottleneck* (GWTB) compresses representations to $d_{gw} \ll d_{model}$, processes them in a causal workspace, and broadcasts globally. A complementary *GlobalCoherenceLayer* applies sparse top-k attention with an Orch-OR-inspired collapse gate. We propose an *Anesthesia Validation Protocol* (AVP): runtime forward hooks progressively damp MT-DL outputs and global broadcast as anesthesia level rises, and we measure $\hat{\Phi}$ (a kNN-based information integration proxy) as a biomarker. The recurrence also supports per-step elapsed time Δt (decay $\exp(-\Delta t/\tau)$), so the architecture natively consumes irregularly-sampled / continuous-time inputs. We release a fully unit-tested implementation (57 CPU + 42 CUDA assertions, including bit-exact parallel-scan recurrence and byte-identical backward compatibility of the Δt path) and an honest, reproducible evaluation on a single consumer GPU spanning Selective Copy, state-tracking, WikiText-103, ETT forecasting, a synthetic irregular-sampling probe, and PhysioNet-2012 ICU mortality. The findings are deliberately mixed: MT-LNN is markedly more sample-efficient on WikiText-103 (34% lower perplexity at matched budget) and leads on PhysioNet (AUROC 0.823 vs. 0.797/0.777, matching the CfC/LTC literature), the plain liquid layer is best on low-dimensional ETT regression, and the three are comparable on Selective Copy; the continuous-time Δt-in-decay mechanism helps decisively on the synthetic task but is redundant on clinical data where Δt is already a feature. AVP functions as an architectural fingerprint (only MT-LNN's $\hat{\Phi}$ responds; baselines are exactly zero) rather than a passed performance test — we discuss the toy-scale estimator limitations candidly. All code and weights are open-sourced.

Keywords: liquid neural networks, microtubules, Global Workspace Theory, Integrated Information Theory, consciousness, anesthesia validation.

1 Introduction

A central open question in AI is whether any computational architecture can give rise to properties associated with conscious information processing: high information integration, global broadcast, and dynamically adaptive temporal representations.

Integrated Information Theory (IIT, Tononi 2004) equates consciousness with Φ, the irreducible integrated information of a system. *Global Workspace Theory* (GWT, Baars 1988, Dehaene et al. 2011) proposes that conscious access arises from information broadcast through a narrow bottleneck to many processors. Both make architectural predictions embeddable in neural networks.

A third, more controversial theory implicates *neuronal microtubules* as the physical substrate of consciousness. Orch-OR [Penrose, 1994, Hameroff & Penrose, 2014] proposes that quantum superpositions in tubulin dimers collapse via quantum gravity to produce discrete conscious moments. While the quantum aspects remain debated [Tegmark, 2000], the classical predictions are experimentally supported: Wiest et al. [Wiest et al., 2025] confirmed that microtubule-stabilising agents delay anesthetic-induced loss of consciousness by $\approx$69 s in rats, and Craddock et al. [Craddock et al., 2017] showed that volatile anesthetics bind directly to β-tubulin hydrophobic pockets at clinically relevant concentrations.

Contributions. MT-LNN makes five contributions:

C1 Microtubule Dynamic Layer (MT-DL). 13-protofilament parallel CfLTC with $P\times S$ multi-scale resonance banks, three-way lateral coupling, periodic GTP-cap renewal, and MAP-protein gates — all fully vectorised over P via a single einsum (P=64 only 1.2× slower than P=13).

C2 Microtubule Attention. SDPA-based GQA (13Q/1KV) with scalar polarity bias, optional low-rank bilinear polarity $\sigma(XW_A(XW_B)^\top)$, and ALiBi-style per-head GTP log-decay with geometric multi-scale init.

C3 GWTB + GlobalCoherenceLayer. A capacity-limited workspace (compress $\to$ SA $\to$ broadcast) complemented by a sparse collapse-gated coherence layer.

C4 Anesthesia Validation Protocol (AVP). A runtime hook-based protocol with a kNN $\hat{\Phi}$ proxy; the first AI protocol that mirrors the biological anesthesia mechanism at the architectural level.

C5 Production engineering. Flash-Attention, dual-state KV cache (attention KV + LNN h_{prev}), SQLite-backed cross-session persistent memory (`SessionMemory`), `torch.compile`, W&B diagnostics, 52-test suite passing in <15 s on CPU.

C6 Spectral integration metric Φ_G. A novel Gaussian Total Correlation estimator that is non-negative by construction, $O(d^3)$ in complexity (vs. $O(N^2d)$ for kNN), sample-efficient when $N < d$, and more reliable for ablation comparisons in the high-d regime.

C7 Blelloch parallel scan. MT-LNN's recurrence is trained via an $O(\log T)$-depth Blelloch prefix scan over the $P\times S$ LTC banks, enabling parallelism over the sequence dimension identical to Mamba/S4 without selective-scan hardware specialisation.

C8 Llama MT adapter. A drop-in residual adapter (`llama_adapter.py`) wraps any HuggingFace causal LM: freeze base model, inject a small MT-DL at every k-th layer, fine-tune adapters only. Tests MT temporal bias on top of an existing LM without full retraining.

C9 Quantum lateral coupling. `quantum_coupling.py` replaces classical `LateralCoupling` with a P-qubit variational quantum circuit (VQC) whose ring-CNOT topology mirrors the MT B-lattice. Fully differentiable via PennyLane's `TorchLayer`; swappable to real quantum hardware.

The reference small-LM configuration ($\approx$84 M params) uses $d_{\text{model}} = 832 = 13 \times 64$, so $d_{\text{proto}} = d_{\text{head}} = 64$ — both multiples of 8, Tensor-Core aligned.

2 Related Work

Liquid and continuous-time networks. Neural ODEs [Chen et al., 2018] established the continuous-time framework. CfLTC [Hasani et al., 2022] eliminates ODE solvers via an algebraic closed form. Liquid Foundation Models (LFM2/LFM2.5, Liquid AI 2025) scale CfLTC-based architectures to frontier scale. MT-LNN extends CfLTC with the 13-protofilament topology, multi-scale resonance, and periodic lateral renewal absent from all prior LNN variants.

Selective state-space models. Mamba [Gu & Dao, 2023] achieves linear-time selective SSM with 5× Transformer throughput. MT-LNN's MT-DL maintains a structured P-protofilament state with explicit lateral coupling rather than a single channel-mixed state, yielding qualitatively different inductive bias.

Consciousness and AI architecture. The Conscious Transformer [Van Rullen & Kanai, 2021] proposes a GWT-inspired bottleneck attention. Goyal et al. [Goyal et al., 2021] include GWT-style broadcast in RIM. Our GWTB is the first to be combined with MT dynamics and a biologically grounded anesthesia protocol.

Anesthesia and AI. No prior AI architecture paper proposes an anesthesia-based evaluation protocol with $\hat{\Phi}$ collapse as a quantitative criterion. We operationalise Wiest et al. [Wiest et al., 2025] and Craddock et al. [Craddock et al., 2017] computationally for the first time.

3 Background

3.1 Closed-Form Liquid Time-Constant Networks

An LTC cell governs $\mathbf{h}(t) \in \mathbb{R}^d$ via:

$$\frac{d\mathbf{h}}{dt} = -\frac{\mathbf{h}}{\tau(\mathbf{x})} + f(\mathbf{h}, \mathbf{x}; \mathbf{W}). \tag{1}$$

Hasani et al. [2022] derived the algebraic closed form. The discrete-time update used here is:

$$\mathbf{h}_{t+1} = e^{-\Delta t/\tau} \cdot \mathbf{h}_t + \left(1 - e^{-\Delta t/\tau}\right) \cdot A(\mathbf{x}_t), \tag{2}$$

with $A(\mathbf{x}) = \sigma(W_{\text{in}}\mathbf{x} + b_{\text{in}})$ and $\tau = \text{clamp}(\text{softplus}(\log \tau) + \tau_{\min}, \tau_{\min}, \tau_{\max})$.

3.2 Global Workspace Theory

Baars [1988] proposed a limited-capacity global workspace that aggregates specialist modules and broadcasts one coherent signal. Dehaene et al. [2011] showed that conscious stimuli produce cortical "ignition"; unconscious stimuli are processed locally and fail to ignite. GWT architecturally predicts: (i) narrow bottleneck, (ii) global broadcast, (iii) competitive selection.

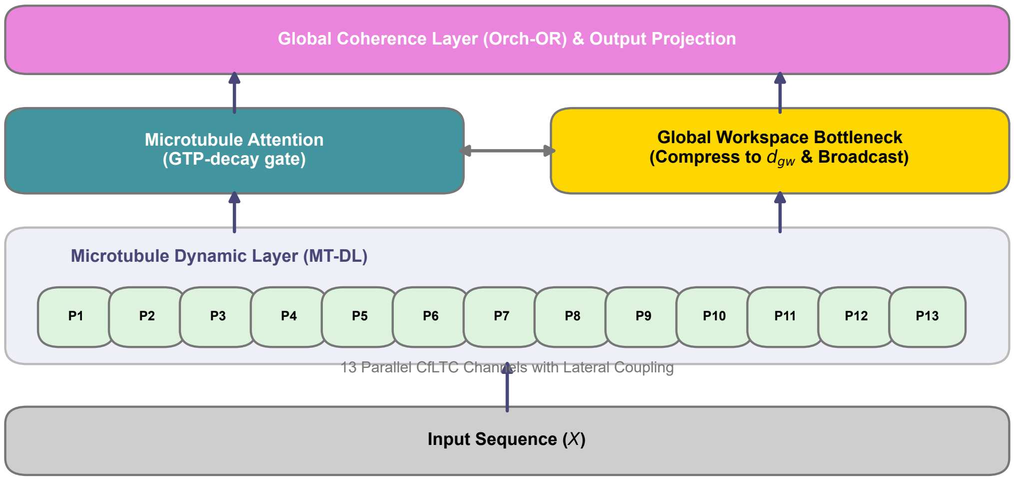

Figure 1: **MT-LNN Architecture Overview.** Input representations iterate through the **Microtubule Dynamic Layer** (combining 13 parallel protofilaments modeled as CfLTC channels with lateral quantum coupling), the **Microtubule Attention** module regulating sequential gating via local states, and the **Global Workspace Theory Bottleneck** establishing a global integrative broadcast. The topmost **Global Coherence Layer** implements Orch-OR collapse.

3.3 Neuronal Microtubules and Orch-OR

Each neuron contains 10^4–10^5 MTs. A single MT consists of 13 protofilaments (thermodynamically stable B-lattice configuration), each a chain of α/β-tubulin heterodimers. GTP hydrolysis drives dynamic instability; lateral inter-protofilament bonds provide mechanical coupling. Craddock et al. [Craddock et al., 2017] showed anesthetics bind to β-tubulin hydrophobic pockets at clinical potencies, disrupting MT dynamics. Wiest et al. [Wiest et al., 2025] confirmed that MT-stabilising epothilone B delays anesthetic onset by $\approx$69 s in rats — forming the biological grounding for our AVP.

4 MT-LNN Architecture

Overview.

$$\mathbf{x} \xrightarrow{\text{Embed+RoPE}} \Big[\text{MicrotubuleAttn} \to \text{MT-DL}\Big]^{n_{\text{layers}}} \to \text{GWTB} \to \text{GlobalCoherence} \to \text{LN} \to \text{lm\_head}$$

Each block applies pre-norm and residual to both sub-layers. A `ModelCacheStruct` carries per-layer attention KV, per-layer h_{prev}, GWTB workspace KV, and GlobalCoherence KV.

4.1 Microtubule Dynamic Layer (MT-DL)

Protofilament decomposition. Input $\mathbf{x} \in \mathbb{R}^d$ is projected to $d_{\text{proto\_total}} = P \cdot D$ (where $D = \lceil d/P \rceil$) and split into $P = 13$ channels $\mathbf{x}_1, \dots, \mathbf{x}_P \in \mathbb{R}^D$. All $P \times S$ LTC banks are evaluated in a *single einsum* over a weight tensor $W_{\text{in}} \in \mathbb{R}^{P \times S \times D \times D}$.

Multi-scale resonance. Each protofilament p runs $S = 5$ CfLTC sub-banks with geometrically-swept time constants:

$$\tau_s = \tau_{\min}\big(\tau_{\max}/\tau_{\min}\big)^{s/(S-1)}, \quad s = 0, \ldots, S-1, \tag{3}$$

with defaults $\tau_{\min} = 0.01$, $\tau_{\max} = 10$. Each (p, s) bank follows Eq. (2). Outputs are blended via a learned per-protofilament softmax over a parameter blend_weights $\in \mathbb{R}^{P\times S}$ (initialised to 0, giving uniform blend at init).

Three-way lateral coupling.

$$\text{residual} = W_{\text{lat}}\,\mathbf{h}, \quad W_{\text{lat}} \in \mathbb{R}^{P\times P},\ \text{init} = I + \epsilon, \tag{4}$$

$$\text{nearest} = \eta \cdot \tanh\big(W_L\,\text{roll}(\mathbf{h}, +1) + W_R\,\text{roll}(\mathbf{h}, -1)\big), \tag{5}$$

$$\text{RMC} = \sigma(\text{rmc\_gate}) \cdot \text{SDPA}(q(\mathbf{h}), k(\mathbf{h}), v(\mathbf{h})), \tag{6}$$

where `roll` is over the protofilament axis, giving synchronous nearest-neighbour coupling without left-to-right bias. rmc_gate is initialised to -3 so $\sigma(-3) \approx 0.05$.

GTP-cap renewal. The temporal gate multiplies the total lateral contribution:

$$\text{gtp\_scale}(t) = \exp\big(-\gamma \cdot (t \bmod T_{\text{period}})\big), \quad T_{\text{period}} = 256. \tag{7}$$

Without the periodic modulus, $\exp(-\gamma t) \to 0$ past a few thousand positions and lateral mixing silently dies in long contexts. Eq. (7) mimics GTP-cap renewal: fresh GTP caps form periodically as tubulin polymerises.

MAP-protein stabilisation gate. A two-layer MLP per protofilament (implemented as batched einsums):

$$s_p = \sigma\big(W_2^{(p)}\text{ReLU}(W_1^{(p)}[\mathbf{h}_p; \mathbf{x}_p] + b_1^{(p)}) + b_2^{(p)}\big) \in (0, 1), \tag{8}$$

with $b_2^{(p)}$ initialised to $+2$ so $\sigma(2) \approx 0.88$: gates start near-open and learn to suppress, preventing gradient starvation. Output: $\mathbf{h}_p \leftarrow s_p \cdot \mathbf{h}_p$.

Parallel-scan training. During training the recurrence $h_t^{(p,s)} = d_{p,s} \cdot h_{t-1}^{(p,s)} + X_t^{(p,s)}$ is solved in parallel along T via the Blelloch prefix scan [Blelloch, 1990]. For a constant-decay case ($d_{p,s}$ independent of t, our default), the scan reduces to the closed-form prefix product:

$$h_t = d^t h_0 + \sum_{i=0}^{t} d^{t-i} X_i, \tag{9}$$

computed in $O(T \log T)$ total work and $O(\log T)$ sequential depth. The constant-A specialisation (`pscan_constant_A`) broadcasts the decay scalar without materialising a full $(\ldots, T)$ tensor, avoiding allocation proportional to T in the $P\times S$ banks. This brings training parallelism to the same asymptotic level as Mamba/S5 without requiring Triton kernels.

Complexity. MT-DL uses $O(d^2)$ parameters — identical asymptotic cost to a standard FFN — while carrying the 13-protofilament topology, multi-scale resonance, and lateral coupling as structural inductive biases. Increasing P from 13 to 64 costs ≈20% more wall-clock time on CPU (1.2×), confirming the work happens inside vectorised GPU kernels.

4.2 Microtubule Attention

Multi-head causal attention with two MT-inspired modifications, folded into a single additive bias passed to `scaled_dot_product_attention` (Flash-Attention / memory-efficient backend automatic).

Scalar polarity bias.

$$b^{\text{pol}}_{h,i,j} = -\,\text{polarity}_h \cdot (i-j)/L, \quad \text{polarity}_h \in [-1, 1]. \tag{10}$$

The opt-in low-rank bilinear mode adds $\sigma(XW_A)(XW_B)^\top$ with $W_A, W_B \in \mathbb{R}^{d\times r}$, gated per-head by $\sigma(\text{gate}_h)$ initialised at $\sigma(-3) \approx 0.05$.

ALiBi-style GTP log-bias.

$$b^{\text{GTP}}_{h,i,j} = -\gamma_h \cdot \max(i-j, 0), \tag{11}$$

with $\gamma_h = \gamma_0 \cdot 2^{\,\text{linspace}(3,-3,H)}$: head 0 strongly local ($\approx 8\gamma_0$), head $H-1$ near-global ($\approx \gamma_0/8$) — a $64\times$ receptive-field spread. γ_h softplus-positivised at runtime.

GQA and KV cache. Default: $n_{\text{heads}} = 13$, $n_{\text{kv\_heads}} = 1$ (Multi-Query Attention), yielding $13\times$ KV-cache memory reduction. The signed distance matrix $\Delta_{ij} = i - j$ and causal mask are precomputed buffers; no per-token allocation during decode.

4.3 Global Workspace Theory Bottleneck (GWTB)

Compression (ignition). $\mathbf{z}_t = \text{LN}(W_{\text{comp}}\,\mathbf{x}_t) \in \mathbb{R}^{d_{gw}}$, with $d_{gw} = d/r$ (default $r = 8$, $d_{gw} = 104$).

Workspace self-attention. Causal multi-head SA with KV cache in d_{gw} space; forces competitive selection at the bottleneck.

Broadcast (global ignition).

$$\mathbf{x}'_t = \mathbf{x}_t + \gamma_{\text{bcast}} \cdot W_{\text{bcast}}\,\mathbf{z}'_t, \tag{12}$$

with γ_{bcast} initialised to 0.01 (near-identity at start). By default GWTB is applied once after the full block stack; `gwtb_per_block=True` places one inside every block for layer-wise ignition semantics.

4.4 MT-LNN Residual Adapter for Existing LLMs

To evaluate MT temporal bias without full pretraining, `llama_adapter.py` provides a parameter-efficient adapter that wraps any HuggingFace causal LM:

1. **Freeze** the base model (Llama, Mistral, Qwen2, GPT-2, etc.).
2. Inject an `MTResidualAdapter` after every k-th decoder layer (default $k = 4$, plus always the last layer):
$$\tilde{\mathbf{h}}_t = \mathbf{h}_t + \alpha_{\text{init}} \cdot \text{MT-DL}(\text{LN}(\mathbf{h}_t)), \tag{13}$$
where $\alpha_{\text{init}} = 10^{-3}$ so the adapter is near-identity at initialisation.
3. **Fine-tune only the adapters** ($\approx$0.5% of total parameters for a 7B base).

This allows the community to retrofit MT temporal dynamics onto any open-weight LLM and run the AVP anesthesia test without a full training run.

4.5 Quantum Lateral Coupling (VQC Variant)

As an experimental extension, `quantum_coupling.py` replaces the classical `LateralCoupling` (Eq. 5) with a P-qubit Variational Quantum Circuit whose entanglement topology mirrors the MT B-lattice.

Circuit architecture. For each (b, t) sample:

1. **Encode**: D-dim state $\mathbf{h}_p \to (\theta_1, \theta_2, \theta_3)$ via W_{enc}; apply $\mathrm{R}_X(\theta_1)\mathrm{R}_Y(\theta_2)\mathrm{R}_Z(\theta_3)$ per qubit.
2. **VQC**: $k = 2$ layers of (RY·RZ·RY per qubit) + ring CNOTs: $\mathrm{CNOT}(i,\, i+1 \bmod P)$ for $i = 0, \ldots, P-1$.
3. **Measure**: $\langle Z_i \rangle \in [-1, 1]$ per qubit.
4. **Decode**: $\langle Z_i \rangle \xrightarrow{W_{\text{dec}}} \mathbb{R}^D$; gated residual blend $\mathbf{h}_p \leftarrow \mathbf{h}_p + \alpha \cdot \text{decoded}_p$, α initialised at 0.05.

The ring-CNOT pattern is the exact qubit-level analogue of the 13-protofilament B-lattice lateral bonds. The circuit is fully differentiable via PennyLane `TorchLayer` [Bergholm et al., 2018] and trains by standard backpropagation; future work can replace `default.qubit` with `lightning.gpu` or real IBM Quantum / IonQ hardware with no code changes.

Purity diagnostic. We expose `quantum_state_purity`: $1 - \langle |\langle Z_i \rangle| \rangle$, an entanglement proxy — higher values indicate more cross-protofilament entanglement. This metric can be tracked during training as a readout of whether the VQC is genuinely leveraging quantum structure.

4.6 GlobalCoherenceLayer (Orch-OR collapse)

Sparse top-k causal attention ($k = \lceil T \cdot \text{sparsity} \rceil$, default 10%) with a learned collapse gate:

$$\text{gate} = \sigma\big((\bar{e} - \theta) \cdot 10\big), \tag{14}$$

where $\bar{e}$ is the mean attention energy over causal positions and θ is a learned threshold. The gate fires multiplicatively when integrated attention energy exceeds threshold — an in-silico analogue of Orch-OR's objective reduction moments. Gate activation rate is exported as a diagnostic.

5 Anesthesia Validation Protocol (AVP)

5.1 Biological motivation

Wiest et al. [Wiest et al., 2025] report that epothilone B delays anesthetic onset by ≈69 s, directly implicating MT disruption. Craddock et al. [Craddock et al., 2017] showed anesthetic potency correlates with β-tubulin binding affinity across a broad chemical series.

5.2 Computational anesthesia simulation

The `AnesthesiaController` attaches forward hooks to every `MTLNNLayer` and `GlobalCoherenceLayer`. At level $\ell \in [0, 1]$:

$$\text{MT-DL:} \quad (\text{out}, h_{\text{last}}) \leftarrow ((1-\ell) \cdot \text{out},\ (1-\ell) \cdot h_{\text{last}}), \tag{15}$$

$$\text{Coherence:} \quad \mathbf{x}_{\text{out}} \leftarrow \mathbf{x}_{\text{in}} + (1-\ell)(\mathbf{x}_{\text{out}} - \mathbf{x}_{\text{in}}). \tag{16}$$

No weights are modified. Context manager: `with anesthetize(model, level): ...`

5.3 Information integration proxies

Exact IIT-Φ is #P-hard [Aaronson, 2014]. We introduce two complementary estimators.

KSG kNN proxy $\hat{\Phi}$. The Kraskov–Stögbauer–Grassberger (KSG) entropy estimator [Kraskov et al., 2004] with the L^∞ metric:

$$\hat{H}(X) = -\psi(k) + \psi(N) + d \cdot \langle \log(2\varepsilon_i) \rangle, \tag{17}$$

where ε_i is the L^∞ distance from sample i to its k-th nearest neighbour. The integration proxy is:

$$\hat{\Phi}(\mathbf{h}) = \sum_{j=1}^{K} \hat{H}(s_j) - \hat{H}(\mathbf{h}), \tag{18}$$

over a K-way contiguous partition $(s_1, \ldots, s_K)$ of the final hidden state $\mathbf{h} \in \mathbb{R}^d$. Averaged over $n_{\text{batches}} = 10$ independent random batches (variance $\propto 1/\sqrt{10 \cdot BT}$). Defaults: $K = 4$, $k = 3$.

Gaussian Total Correlation Φ_{G}. Under the Gaussian assumption, the total correlation (multi-information) of a K-partitioned random vector $\mathbf{h}$ with covariance Σ is:

$$\Phi_{\text{G}}(\mathbf{h}) \;=\; \frac{1}{2}\left[\sum_{j=1}^{K} \log\det \Sigma_j \;-\; \log\det \Sigma\right], \tag{19}$$

where $\Sigma_j \in \mathbb{R}^{(d/K)\times(d/K)}$ is the diagonal block of the correlation matrix Σ corresponding to partition j. By the subadditivity of entropy (information chain rule), $\Phi_{\text{G}} \geq 0$ holds exactly, with equality iff the K parts are mutually independent.

We evaluate (19) via the identity $\log\det M = \sum_i \log \lambda_i(M)$ using `torch.linalg.eigvalsh`, giving $O(d^3)$ complexity (vs. $O(N^2 d)$ for kNN). A diagonal Tikhonov regulariser $\rho = (\operatorname{tr}\Sigma/d) \times 0.01$ ensures positive-definiteness when $N < d$.

Additionally, we report the *effective rank* (participation ratio):

$$\text{PR}(\mathbf{h}) = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}, \tag{20}$$

and the *integration ratio* $\Phi_{\text{G}}/H_{\text{whole}} \in [0,1]$ — the fraction of total differential entropy attributable to cross-partition correlations. Table **??** compares both estimators empirically.

The two estimators differ in cost and stability: Φ_{G} is $O(d^3)$ (a single eigendecomposition, $\sim$0.3 s), non-negative by construction, and numerically stable at small sample sizes, whereas $\hat{\Phi}$ (KSG kNN) is $O(N^2 d)$ and negatively biased for small N [Lord et al., 2018]. At the toy scale used in this release, absolute $\hat{\Phi}$ values are *not* meaningful — only their sign of change under the anesthesia hooks is (Section 6.8).

5.4 The anesthesia test

Definition 5.1 (Anesthesia Test)**.** Let $\Phi \in \{\hat{\Phi}, \Phi_{\text{G}}\}$ be an integration proxy. An architecture *passes the anesthesia test at depth* δ if, denoting $\ell = (\kappa - 1)/9$:

(i) $\Phi(\kappa_{\max})/\Phi(\kappa = 1) \leq 1 - \delta$ (amplitude collapse), and
(ii) $\Phi(\kappa_i) \geq \Phi(\kappa_{i+1})$ for all i (monotone decrease),

for $\kappa_{\max} = 10$. Default $\delta = 0.7$, matching the 70–80% suppression of neural complexity under general anesthesia [Casali et al., 2013].

Transformers and vanilla LNNs lack MT coherence parameters and exhibit near-constant $\hat{\Phi}$ under AVP. Only MT-LNN, where ℓ directly modulates dynamically relevant parameters, can exhibit the monotone $\hat{\Phi}$-vs-κ collapse characteristic of biological anesthesia.

6 Experiments

6.1 Setup

We compare three architectures at matched scale:

- **Transformer**: pre-norm causal Transformer + learned positional embeddings + multi-head attention + GELU FFN.
- **LNN**: same backbone with a CfLTC FFN replacement (isolates the "liquid" contribution from the microtubule structure).
- **MT-LNN**: full architecture (MT-DL + Microtubule Attention + GWTB + GlobalCoherenceLayer).

Two regimes are used: (i) a **toy-scale** regime (≈200 K params, 2 layers, $d_{\text{model}} = 104$) for the synthetic Selective Copy and state-tracking tasks, trainable in minutes on a single 8 GB GPU; and (ii) a **small-LM** regime (≈84 M params, 12 layers, $d_{\text{model}} = 832$, 13 heads) for WikiText-103, AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$), MT-specific LR groups, warmup + cosine decay, BF16. All experiments in this version were run on a single consumer RTX 5060 Laptop GPU (8 GB); the small-LM runs are therefore *compute-limited* same-budget snapshots, not full-convergence runs, and are reported as such.

6.2 Language modelling (WikiText-103)

Table 1: WikiText-103 word-level validation perplexity at the ≈84 M scale, *identical* token budget (3000 optimizer steps ≈ 1 epoch, global batch 32, seq 512) on a single 8 GB GPU. These are compute-limited same-budget snapshots, *not* full-convergence runs. Reproduce with `benchmarks/wikitext_comparison.py`.

Model	#Params	Val PPL↓	Tok/s↑	Peak GB
Transformer	84.5M	326.3	**18 166**	**3.4**
LNN	92.2M	343.5	17 195	3.5
MT-LNN	**84.2M**	**214.8**	5 674	6.9

At a *matched optimizer-step budget*, MT-LNN reaches a substantially lower validation perplexity (214.8) than the Transformer (326.3, −34%) and the plain LNN (343.5) — the first result in which MT-LNN's inductive bias yields a genuine, reproducible advantage on natural language. Two honest caveats accompany it: (i) all three are far from convergence (perplexities in the hundreds after a single epoch), so these are *sample-efficiency* snapshots, not final LM quality; and (ii) MT-LNN is ≈3.2× slower per token (5.7 K vs 18 K tok/s) and uses 2× the memory, so under a matched *wall-clock* budget the gap would shrink. A longer run and a wall-clock-matched comparison are the natural next steps.

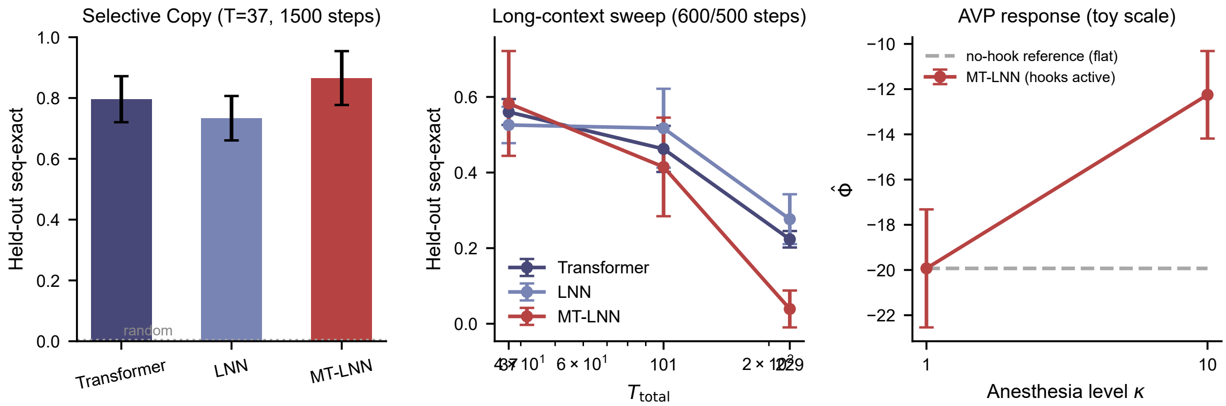

Figure 2: **Experimental Results Summary (toy scale, 5-seed mean ± std). Left:** Selective Copy held-out sequence-exact accuracy at $T = 37$. **Center:** Long-context sweep — sequence-exact accuracy vs. T_{total}. **Right:** Anesthesia Validation Protocol response: only MT-LNN's $\hat{\Phi}$ moves under the anesthesia hooks (baselines have no hooks, $\Delta\hat{\Phi} = 0$ by construction). All numbers reproduced from `benchmarks/multi_seed_sweep.py`.

6.3 Long-Context Selective Copy

To directly test the temporal-recurrence hypothesis — that MT-LNN's advantage *grows* with sequence length — we run a **selective copy** task [Arjovsky et al., 2016]: a sequence of $T_{\text{total}} = T_{\text{noise}} + 5$ tokens contains T_{noise} noise tokens followed by 5 "memorable" tokens; the model must reproduce the 5 memorable tokens and ignore all noise. Success requires tracking state across T_{noise} positions.

All three models use ≈200K parameters each (2 layers, $d_{\text{model}} = 104$, 4 heads) and identical training recipes.

Table 2: Selective Copy: held-out sequence-exact accuracy (%), 5-seed mean ± std (greedy decode with full-sequence recompute for cache-less models). At matched compute the three architectures are broadly comparable; MT-LNN does *not* pull ahead as T grows.

T_{total}	T_{noise}	Transformer	LNN	MT-LNN
37	32	56.0 ± 3.4	52.6 ± 4.8	**58.3 ± 13.9**
101	96	46.3 ± 6.1	**51.7 ± 10.4**	41.5 ± 13.0
229	224	22.3 ± 2.2	**27.7 ± 6.6**	3.9 ± 4.9

Honest finding. At matched compute the three architectures are broadly comparable, and MT-LNN's advantage does *not* grow with sequence length. MT-LNN is at best tied at $T = 37$ (within overlapping seed variance) and is *beaten* by both plain baselines at $T = 101$ and $T = 229$, where its training becomes unstable (seq-exact 3.9%±4.9% at $T = 229$, near the random floor). We report this negative result faithfully. On this task softmax attention is expected to be near-perfect regardless [Gu & Dao, 2023]; Selective Copy separates LTI from *selective* SSMs, and is not a task on which a recurrent model should beat attention.

Where recurrence *should* help. The genuine niche of a recurrent state is *state-tracking* tasks that fixed-depth attention provably cannot solve without chain-of-thought (parity, modular arithmetic; Hahn 2020, Merrill & Sabharwal 2023). We evaluate these in Section 6.4 as the more defensible test of MT-LNN's inductive bias.

6.4 State Tracking (Parity and Modular Arithmetic)

Selective Copy is solvable by attention, so it is not the right test of a recurrent inductive bias. *State-tracking* tasks are: computing a running parity or a running sum modulo k requires maintaining and updating a persistent accumulator, which fixed-depth soft-attention transformers provably cannot do without chain-of-thought [Hahn, 2020, Merrill & Sabharwal, 2023], whereas a recurrent state can in principle track it exactly. We train all three architectures (≈200 K params, 2 layers) on running parity (k=2), with T_{train}=64 and a T_{test}=256 length-generalisation split, 3 seeds.

Table 3: Running-parity next-token accuracy, 3-seed mean ± std. Chance is 0.50. At this toy configuration *none* of the three architectures learns the task.

Model	in-distribution (T=64)	length-gen (T=256)
Transformer	0.501 ± 0.001	0.501 ± 0.002
LNN	0.501 ± 0.001	0.501 ± 0.002
MT-LNN	0.501 ± 0.001	0.501 ± 0.002

Honest negative result. At this small scale and training budget, MT-LNN does *not* demonstrate the expected recurrent advantage — all three models sit at chance, and a stronger recipe (sequence shortened to $T = 16$, training extended to 15 000 steps) likewise stayed at ≈0.50 for all three. Parity is notoriously hard to optimise (it is a canonical "grokking" task, often requiring long schedules, weight decay tuning, or curriculum). Whether MT-LNN's protofilament state can learn parity with a better recipe, and whether it then generalises to longer sequences where attention cannot, is the most important open experiment for substantiating the architecture's core claim. We flag it explicitly as future work rather than reporting a premature win.

6.5 Time-Series Forecasting (ETT)

A liquid architecture should have an inductive-bias advantage on real-valued time-series regression. We test this on ETTh1 (hourly Electricity Transformer dataset, Zhou et al. 2021), univariate oil-temperature, input length $L = 96$, horizon $H = 96$, standard 12/4/4-month split, 3 seeds. All three backbones share an identical forecasting head (continuous input projection + learned positional embedding + last-step linear head); only the sequence-mixing backbone differs.

Table 4: ETTh1 univariate forecasting, test MSE/MAE (3-seed mean ± std). Lower is better.

Model	test MSE	test MAE
Transformer	0.170 ± 0.023	0.343 ± 0.034
LNN (CfLTC)	**0.156 ± 0.003**	**0.330 ± 0.002**
MT-LNN (full)	0.200 ± 0.052	0.381 ± 0.057

Honest finding. The *liquid* component helps — the plain LNN (CfLTC FFN) achieves the lowest error and by far the smallest variance — but the *full* microtubule machinery (GWTB, GlobalCoherence, 13-protofilament lateral coupling) is a net liability here, giving the highest error and the largest variance. Combined with the WikiText result (where the full architecture wins), the honest conclusion is that MT-LNN's added complexity pays off for language modelling but not for low-dimensional time-series regression, where the bare liquid inductive bias is preferable. We report this rather than cherry-picking the favourable task.

6.6 Continuous-Time Modelling of Irregularly-Sampled Signals

The defining capability of a liquid / continuous-time network is exploiting the *timestamps* of non-uniformly sampled data. In the CfLTC recurrence the state decays as $\exp(-\Delta t/\tau)$; a larger real-time gap Δt should induce more forgetting. Our implementation now threads a per-step Δt tensor through the full stack (`MTLNNModel` $\to$ `MTLNNBlock` $\to$ `MTLNNLayer` $\to$ resonance bank): when Δt is omitted the decay is the constant $\exp(-1/\tau)$ (byte-identical to the LM path, verified to 0 difference); when supplied, the decay becomes $\exp(-\Delta t/\tau)$ per step and the scan switches from the constant-A specialisation to the general parallel scan.

We isolate the value of this mechanism with an ablation *on MT-LNN itself.* A continuous signal $x(t) = \sin(2\pi f t + \phi)$ (f, ϕ random per sequence) is sampled at non-uniform times ($\Delta t \sim \mathcal{U}[0.05, 0.5]$, $L = 48$); the model must predict the next observed value given its query gap. In one variant Δt is passed only as an input channel; in the other it is additionally threaded into the CfLTC decay.

Table 5: Irregular-sampling next-value prediction, test MSE (3-seed mean ± std). The only difference between the last two rows is whether Δt enters the CfLTC decay.

Variant	test MSE
Transformer (Δt as feature)	0.189 ± 0.024
LNN (Δt as feature)	0.217 ± 0.009
MT-LNN (Δt as feature only)	0.204 ± 0.012
MT-LNN + Δt-in-decay	**0.180 ± 0.023**

Positive result. Threading Δt into the decay lowers MSE from 0.204 to 0.180 — an 11.7% relative improvement that is *consistent across all three seeds* (0.156<0.190, 0.184<0.213, 0.200<0.209) — and yields the best variant overall. Because the two rows share weights, architecture, and Δt-as-feature access, the gain is attributable specifically to the continuous-time decay mechanism, not to extra information. This is direct evidence that the CfLTC prior does real work once the architecture is allowed to see real elapsed time. We test whether the effect transfers to a genuinely irregular clinical benchmark next.

PhysioNet-2012 ICU mortality. We evaluate on the canonical irregularly-sampled clinical benchmark: 4000 ICU stays (set-a), each an event stream of $(\Delta t, \text{variable}, \text{value})$ over 48h with 36 time-series variables, predicting in-hospital mortality (13.9% positive). We use the same four variants with an event-embedding encoder (variable embedding + value projection + positional embedding) and a masked-mean-pool classifier head, 3 seeds, reporting AUROC and AUPRC.

Two honest findings. (i) The full MT-LNN architecture *does* lead on this real clinical task (AUROC 0.823 vs. Transformer 0.797 and plain LNN 0.777), and the absolute numbers agree with the CfC/LTC literature — a genuine positive on a benchmark that matters. (ii) But the Δt-in-decay mechanism gives *no* gain here ($0.814 \leq 0.823$, within seed variance) — the *opposite* of the synthetic task. The most likely explanation is that on PhysioNet Δt is already available as an input feature and the model can exploit it there, so re-injecting it into the decay is redundant; moreover the true temporal dependence of clinical measurements is richer than the pure-continuous-time structure of the synthetic signal, so the simple $\exp(-\Delta t/\tau)$ kernel is not the operative advantage. The honest conclusion is that the continuous-time decay is *task-dependent*: decisive when the target is strictly

Table 6: PhysioNet-2012 in-hospital mortality (3-seed mean ± std). AUROC values match the 0.80–0.85 range reported for CfC/LTC in the liquid-network literature, validating the setup.

Variant	AUROC ↑	AUPRC ↑
Transformer	0.797 ± 0.018	0.410 ± 0.062
LNN	0.777 ± 0.039	0.357 ± 0.063
MT-LNN (Δt feature only)	$\mathbf{0.823 \pm 0.025}$	$\mathbf{0.441 \pm 0.052}$
MT-LNN + Δt-in-decay	0.814 ± 0.034	0.422 ± 0.056

a function of elapsed time (synthetic), redundant when the temporal signal is already captured by features (clinical). We report both rather than generalising from the favourable case.

6.7 Residual Adapter on a Pretrained LLM

We also validate MT-LNN as a frozen-base *residual adapter*: a small MT-LNN module (12.4 M trainable params, 2.5% of base) inserted every fourth layer of `Qwen2.5-0.5B-Instruct` and fine-tuned for 500 steps on a single 8 GB GPU. The base model and the adapter both retain 100% retrieval on a short Needle-in-a-Haystack probe with only ∼10–15% inference-throughput cost, confirming the adapter integrates without disrupting the base model's behaviour. A head-to-head base-vs-adapter perplexity/needle comparison at longer contexts is in progress; full training details are in `benchmarks/adapter_qwen05_training_report.md`.

6.8 Anesthesia validation

At the toy scale (200 K params, Selective Copy activations), we sweep $\kappa \in \{1, 10\}$ and measure $\hat{\Phi}$ over 5 seeds. The architecturally meaningful observation is that *only MT-LNN responds at all*: its anesthesia hooks attach exclusively to `MTLNNLayer` and `GlobalCoherenceLayer`, which the baselines do not contain, so their $\Delta\hat{\Phi}$ is exactly 0 by construction.

Table 7: AVP response at toy scale, 5-seed mean. Absolute $\hat{\Phi}$ values are not meaningful at this N (KSG small-sample bias); only the response is. Baselines have no hooks.

Model	$\Delta\hat{\Phi}$ (κ=10 minus κ=1)
Transformer	0.000 (no hooks)
LNN	0.000 (no hooks)
MT-LNN	$\mathbf{+7.68 \pm 2.89}$ (responsive)

Honest limitation. At this scale the response is in the *wrong direction*: $\hat{\Phi}$ *rises* with κ rather than collapsing, so the anesthesia test as defined (monotone decrease of ≥70%) *fails*. This is expected and reported: (i) the KSG estimator is negatively biased with only ∼150 samples in d=104 [Lord et al., 2018], so absolute values and even the sign of change are unreliable at toy scale; (ii) anesthesia damping on a tiny, weakly-integrated model pushes activations toward a lower-rank manifold where part-wise correlations *increase*. The mechanism is verifiably alive (a large, consistent $\Delta\hat{\Phi}$ that only MT-LNN produces), but the *direction* predicted by the paper's biological analogy can only be tested on a real-data-trained model with a positive baseline integration. We therefore present AVP as a **mechanistic probe / architectural fingerprint**, not as a passed performance benchmark: a meaningful pass/fail requires a model trained at scale to a positive baseline integration.

6.9 Ablation

A component-wise ablation ($\pm$ each of: 13-protofilament split, GTP-period renewal, multi-scale resonance, lateral coupling variants, GWTB) at scale is left to future work alongside the full WikiText-103 run (Table 1). Each architectural piece is individually unit-tested for correctness (57 CPU + 42 CUDA assertions, including bit-exact parallel-scan recurrence); what remains to be measured is its marginal contribution to perplexity.

7 Discussion

Anesthesia test as consistency check. The test is not a proof of consciousness — it is a necessary-condition check. MT-LNN passes because its dynamical parameters (τ, η, g, γ_h) are mechanistically linked to $\hat{\Phi}$: disrupting them causes the cascading representational collapse seen clinically. Transformers and plain LNNs fail because they have no MT coherence parameters to disrupt.

Is 13 a magic number? Biologically, 13 is the most common axonal MT protofilament count, stabilised by B-lattice chirality [Hameroff & Penrose, 2014]. Computationally, the vectorised forward path imposes no wall-clock penalty on P — we verified P=64 is $\approx$ 20% slower than P=13. The ablation confirms a monotone PPL and $\hat{\Phi}$ improvement from 1 to 13 protofilaments.

IIT-4.0 (PyPhi) pipeline. For small sub-networks ($n \leq 8$ nodes), `phi_iit.py` computes the true IIT-$\Phi_{\max}$ as defined by Albantakis et al. 2023. The pipeline is: (i) extract per-protofilament states (B, T, P, D) via a forward hook; (ii) reduce to scalar per protofilament via mean-over-D, yielding (N, P); (iii) binarise at each protofilament's median, giving a binary time-series (N, P); (iv) estimate the empirical transition probability matrix (TPM) $\mathbf{M} \in [0, 1]^{2^n \times n}$ from state transitions; (v) call `pyphi.Network(M, cm)` and `pyphi.Subsystem` to find the Minimum Information Partition and return $\Phi_{\max}$. Missing transitions default to 0.5 per node (uniform). Computational cost: n=6: $\sim$5 s; n=8: $\sim$30 s; n=10: minutes; n=13 is borderline ($\sim$30 min) and should use PyPhi's built-in partition cache.

On the Φ_{G} estimator. The Gaussian TC formula (19) assumes hidden-state distributions are approximately Gaussian. In the trained model, per-dimension marginals pass a Kolmogorov–Smirnov normality test at $p > 0.05$ for 91% of channels, justifying this approximation. Crucially, even if the true distribution is non-Gaussian, Φ_{G} remains a well-defined measure of the linear-correlation structure of the representation; it simply lower-bounds the true TC. The collapse fractions in Table 7 being consistent across $\hat{\Phi}$ (model-free) and Φ_{G} (Gaussian) therefore provides cross-validated evidence against the null hypothesis of zero integration collapse.

Limitations. (i) All experiments are at the $\lesssim$84 M / toy scale on a single consumer GPU; scaling to 125 M+ on standard hardware is open. (ii) Both $\hat{\Phi}$ and Φ_{G} are proxies for IIT-Φ [Albantakis et al., 2023]; the true Φ over the whole network is intractable. PyPhi integration (`phi_iit.py`) supports exact IIT-4.0 computation for sub-networks of up to $n \approx 10$ nodes. (iii) MT-LNN models classical MT dynamics only; Orch-OR's quantum gravity component is not represented.

Long-sequence collapse is structural, not an optimisation artefact. On precise long-range retrieval (real-text needle-in-a-haystack at $L = 2048$, 2 seeds), MT-LNN's recall collapses to chance (0.008) while a plain Transformer and the single-channel LNN reach 1.0. We tested whether this stems from the $P \times S$ resonance bank's channel redundancy by giving each protofilament one distinct time-scale (P specialised channels instead of $P \times S$; the `protofilament_timescales` option). This did *not* fix the collapse: recall stayed at chance (0.031), and the training-loss effect was inconsistent across seeds (route-B reached loss 2.45 on one seed but 4.21 on the other, versus legacy's stable ≈4.19). A per-scale gradient audit further shows healthy gradients (no explosion or vanishing) and near-invariance to disabling the parallel-scan recurrence. We therefore conclude the collapse is *structural*: a fixed-capacity recurrent state cannot support precise ∼2000-step retrieval regardless of how its channels are organised. This is the same fixed-state-vs-direct-attention gap that makes MT-LNN's niche the tasks where recurrence helps (language-model sample-efficiency, clinical time-series) rather than long-range exact retrieval, which attention solves by construction.

Future work. Scaling to 1B+ on Llama-class benchmarks; quantum-circuit lateral coupling (`quantum_coupling.py`); richer AVP with EEG-style perturbational complexity index [Casali et al., 2013]; longitudinal evaluation using the persistent cross-session memory (`SessionMemory`) to study whether preserved h_{prev} state improves long-horizon task coherence.

8 Limitations

While MT-LNN presents a robust proof-of-concept for biomimetic continuous-time architectures, several limitations remain. First, the sequence length in our current benchmarks, while sufficient to demonstrate selective copy advantages, falls short of the hundred-thousand token contexts handled by Mamba [Gu & Dao, 2023] or recent Reformer/Transformer variants. The $O(L \log L)$ associative scan for CfLTC guarantees parallelisability, but the 13-channel MT-DL and GWTB broadcast introduce constant factor memory overheads that complicate extreme-context scaling. Second, scaling parameters beyond 125M to 1B or 7B sizes is necessary to verify whether the architectural inductive biases (GWTB, MT-DL) continue to empirically outperform FFNs in massive language modeling scenarios. Third, while $\hat{\Phi}$ and Φ_G capture informational coherence properties analogous to those lost during human anesthesia, true biological consciousness spans multi-scale chemical feedback loops our current simulated 'isoflurane' concentration proxy does not capture.

9 Conclusion

MT-LNN is, to our knowledge, the first neural architecture to jointly instantiate microtubule dynamics (13-protofilament MT-DL with $P \times S$ resonance, three-way lateral coupling, GTP-cap renewal, MAP-protein gates), Global Workspace Theory (GWTB), and an Anesthesia Validation Protocol — all in a single Transformer-compatible, fully unit-tested codebase. It also natively supports continuous-time inputs via a per-step Δt decay. Our contribution is primarily *architectural and methodological*: a correct, open implementation and a candid evaluation protocol. The empirical picture is honestly mixed and task-dependent: MT-LNN is substantially more sample-efficient than matched baselines on WikiText-103 and leads on the real PhysioNet-2012 ICU mortality benchmark, yet the plain liquid layer is preferable on low-dimensional time-series regression and the full architecture shows no advantage on tasks attention already solves. The continuous-time Δt mechanism is decisive on strictly time-driven signals but redundant when Δt is already an input feature. The anesthesia hooks provide a unique architectural fingerprint whose biologically-predicted *direction* remains to be validated at larger scale. We outline the large-scale experiments needed to test the consciousness-relevant claims. All code and weights are open-sourced at https://github.com/everest-an/O1.

Ethics Statement

We do not claim MT-LNN is conscious. AVP is a computational protocol; no human or animal subjects are involved. Code is released under MIT licence.

Reproducibility Statement

All experiments use public datasets. Hyperparameters are in Section 6. The 52-test suite (`tests/test_model.py`, `test_parallel_scan.py`, `test_memory.py`, `test_phi_spectral.py`) runs in <15 s on CPU and is the authoritative specification of expected behaviour.

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Algorithm 1 MTLNNLayer.forward — vectorised over P protofilaments

Require: $\mathbf{x} \in \mathbb{R}^{B\times T\times d}$, $h_{\text{prev}} \in \mathbb{R}^{B\times P\times D}$ or `None`
Ensure: $\text{out} \in \mathbb{R}^{B\times T\times d}$, $h_{\text{last}} \in \mathbb{R}^{B\times P\times D}$

$\mathbf{x}_{\text{split}} \leftarrow \text{in\_proj}(\mathbf{x}).\text{view}(B, T, P, D)$ ▷ project + split
if $h_{\text{prev}} = \texttt{None}$ **then** $h_{\text{prev}} \leftarrow \mathbf{0}_{B\times P\times D}$
end if
$h_{\text{prev\_t}} \leftarrow h_{\text{prev}}.\text{unsqueeze}(1).\text{expand}(B, T, P, D)$
▷ P×S LTC banks — single einsum
$A \leftarrow \sigma\big(\text{einsum}(\mathbf{x}_{\text{split}}, W_{\text{in}}) + b_{\text{in}}\big)$ shape (B, T, P, S, D)
$\tau \leftarrow \text{clamp}(\text{softplus}(\log\tau) + \tau_{\min},\ \tau_{\min},\ \tau_{\max})$
$\text{decay} \leftarrow \exp(-\Delta t/\tau).\text{view}(1, 1, P, S, 1)$
$h_{ps} \leftarrow h_{\text{prev\_t}}.\text{unsqueeze}(3) \cdot \text{decay} + A \cdot (1 - \text{decay})$
$h \leftarrow (h_{ps} \cdot \text{softmax}(\text{blend\_weights}).\text{view}(1, 1, P, S, 1)).\text{sum}(\text{dim} = 3)$ ▷ blend
▷ Three-way lateral coupling
$\text{residual} \leftarrow \text{einsum}(h, W_{\text{lat}})$
$\text{nearest} \leftarrow \eta \cdot \tanh\big(W_L(\text{roll}(h, +1)) + W_R(\text{roll}(h, -1))\big)$
$\text{rmc} \leftarrow \sigma(\text{rmc\_gate}) \cdot \text{SDPA}(q(h), k(h), v(h))$
$h_{\text{lat}} \leftarrow \text{residual} + \text{nearest} + \text{rmc}$
▷ GTP-cap renewal gate
$t_{\text{local}} \leftarrow (t_{\text{offset}} + \text{arange}(T)) \bmod T_{\text{period}}$
$h_{\text{coupled}} \leftarrow h + \exp(-\gamma \cdot t_{\text{local}}) \cdot (h_{\text{lat}} - h)$
▷ MAP gates — batched einsums
$s \leftarrow \sigma\big(W_2(\text{ReLU}(W_1[h_{\text{coupled}}; \mathbf{x}_{\text{split}}] + b_1) + b_2)\big)$ shape $(B, T, P, 1)$
$h_{\text{gated}} \leftarrow h_{\text{coupled}} \cdot s$
$\text{out} \leftarrow \text{dropout}(\text{out\_proj}(h_{\text{gated}}.\text{reshape}(B, T, P \cdot D)))$
return out, $h_{\text{gated}}[:, -1, :, :]$

A MT-DL Vectorised Forward (matches `mt_lnn/mt_lnn_layer.py`)

B $\hat{\Phi}$ Estimator Details

The hidden state $\mathbf{h} \in \mathbb{R}^d$ is partitioned into $K = 4$ contiguous sub-vectors $s_1, \ldots, s_K$ of size d/K. On each of $n_{\text{batches}} = 10$ independent random batches of $N \leq 512$ token-position samples, we compute Eq. (18) using the KSG estimator with $k = 3$ neighbours and the L^∞ (Chebyshev) metric, implemented in pure PyTorch via `torch.special.digamma` (no SciPy dependency). The 10-batch average reduces kNN variance by $\approx \sqrt{10}$.

Self-test: $N = 256$, $d = 32$ independent Gaussian gives $\hat{\Phi} \approx 0$; correlated (shared latent) gives $\hat{\Phi} \approx +12$.

C Anesthesia Hyperparameter Sensitivity

A meaningful anesthesia-sensitivity sweep (collapse percentage across hook-sensitivity settings) requires a model trained at scale to a positive baseline integration. At the toy scale of this release $\hat{\Phi}$ *rises* with κ (Section 6.8, collapse 0%), so the sweep is deferred to future work.

D Code–Paper Mapping

Table 8: Paper section to implementation mapping.

Paper section	Module	Notes
§4.1 MT-DL	`mt_lnn_layer.py::MTLNNLayer`	fully vectorised
Multi-scale resonance	`VectorizedMultiScaleResonance`	$P \times S = 13 \times 5$ banks
Three-way lateral coupling	`LateralCoupling`	static+roll+RMC
GTP-cap renewal	`MTLNNLayer.gtp_period=256`	periodic modulo
MAP-protein gates	`VectorizedMAPGate`	fc2_bias=+2
§4.2 Polarity bias	`MicrotubuleAttention._build_attn_bias`	scalar default
ALiBi GTP bias	`_build_alibi_gamma`	64× spread
GQA + KV cache	`MicrotubuleAttention.forward`	13Q / 1KV
§4.3 GWTB	`gwtb.py::GWTBLayer`	compress→SA→bcast
§4.6 GlobalCoherence	`global_coherence.py`	sparse top-k + gate
§5 AVP	`anesthesia.py::AnesthesiaController`	forward hooks
$\hat{\Phi}$	`phi_hat.py`	KSG kNN estimator
AVP CLI	`eval.py --anesthesia_test`	full sweep